Chapter 3
The Quantum Formalism for Spin

Learning goals:

(1) Describe how a spin formalism (operators and states) can be developed as a consequence of the Stern-Gerlach experiments.
(2) Use Pauli spin-matrix identities in solving problems (generalized Euler identity, product law for two spin matrices, etc.).
(3) Solve eigenvalue problems for two-state models.

Material to review:

(a) The Stern-Gerlach experiment.
(b) The quadratic equation.
(c) Matrix multiplication.
(d) Levi-Cevita antisymmetric tensor.
(e) Complex arithmetic and complex conjugation.

In this chapter, we introduce a formal way to work with quantum mechanics in terms of quantum states and quantum operators. We also describe how the states can be represented in terms of concrete vectors and the operators in terms of concrete matrices. The spin matrices were first developed by Wolfgang Pauli, and they satisfy a number of important identities that we discuss as well. If you are not so familiar with matrices, we give you a short primer on that too. We start to develop the quantum formalism that allows us to express quantum phenomena in a mathematical language which can be used to make predictions for experiments.

3.1 Spin operators from the Stern-Gerlach experiment

One of the critical facts about the Stern-Gerlach experiment was that no matter how we oriented the magnetic field of the analyzer, we always measured two separations and they always were of the same size. If we want to create a quantum state that differentiates between the two, we can call the one that deflects upwards (for a z-axis oriented analyzer) as $|\uparrow\rangle_z$ and the one that deflects downwards as $|\downarrow\rangle_z$. But, we may want to do more than just work with the states, we would like to examine operators, which map states to states.

The simplest mapping we can have is to map to a numerical value related to the splitting in the experiment. We define an operator $\hat{S}_z$ which does the following:

$$\hat{S}_z |\uparrow\rangle_z = \frac{\hbar}{2} |\uparrow\rangle_z \quad \text{and} \quad \hat{S}_z |\downarrow\rangle_z = -\frac{\hbar}{2} |\downarrow\rangle_z . \tag{3.1}$$

Now, we would like to say that the splitting between the two states corresponds to the difference, or $\hbar$, but the experimental value is twice that. This factor of nearly 2 is called the g-factor, and we will ignore it in all of our analysis here, to focus on the spin operators which we are developing. It is simple to add it back in later, when one wants to describe an actual Stern-Gerlach experiment.

The next type of operator we can have is one that maps an up to a down and *vice versa*. Here, we have a couple different choices we can use. We decide to define the spin raising operator $\hat{S}_+$ and the spin lowering operator $\hat{S}_-$ as follows:

$$\hat{S}_+ |\uparrow\rangle_z = 0 \quad \text{and} \quad \hat{S}_+ |\downarrow\rangle_z = \hbar |\uparrow\rangle_z ; \tag{3.2}$$

$$\hat{S}_- |\uparrow\rangle_z = \hbar |\downarrow\rangle_z \quad \text{and} \quad \hat{S}_- |\downarrow\rangle_z = 0. \tag{3.3}$$

That is the raising operator raises a down to an up, but cannot raise an up further, and similarly for the lowering operator. The notion of an operator that annihilates a state turns out to be a very important concept in much of quantum mechanics. We will use this idea many times. The choice of $\hbar$ for the coefficient is made with 20-20 hindsight, as you will soon see. The justification for this value is that it is the difference of the two eigenvalues of $\hat{S}_z$.

Because there are only two possible states along any axis direction for spin, when we propose an operator, we need only show how it acts on the two basis vectors to fully define the operator. But, now that we have done this, we arrive at a curious fact. The operators do not commute. That is, if we apply a $\hat{S}_+$ before a $\hat{S}_z$ the end result is not the same as applying them in the opposite order. Of course, this can happen, but it is clearly something weird that we are not used to. Our first point of business then is to examine this further and see if we can determine what the commutators of these operators are with each other. We do this by applying the operators in different orders to each of the basis vectors and take their difference. We then map this action to an operator we already know. Some examples will make this procedure clearer.

We start with the commutator $[\hat{S}_z, \hat{S}_+] = \hat{S}_z\hat{S}_+ - \hat{S}_+\hat{S}_z$. We need to act this onto $|\uparrow\rangle_z$ and $|\downarrow\rangle_z$. We find the following:

$$\left(\hat{S}_z\hat{S}_+ - \hat{S}_+\hat{S}_z\right) |\uparrow\rangle_z = -\hat{S}_+\frac{\hbar}{2} |\uparrow\rangle_z = 0 \tag{3.4}$$

and

$$\left(\hat{S}_z\hat{S}_+ - \hat{S}_+\hat{S}_z\right)|{\downarrow}\rangle_z = \hat{S}_z\hbar\,|{\uparrow}\rangle_z - \hat{S}_+\left(-\frac{\hbar}{2}\right)|{\downarrow}\rangle_z = \hbar^2\,|{\uparrow}\rangle_z\,. \tag{3.5}$$

What operator do we know that annihilates $|{\uparrow}\rangle_z$ and transforms $|{\downarrow}\rangle_z$ to $|{\uparrow}\rangle_z$? It is $\hat{S}_+$, and by comparing the constants, we find that we have just derived

$$[\hat{S}_z, \hat{S}_+] = \hbar\hat{S}_+. \tag{3.6}$$

Note that this is a relationship between operators that we discovered by acting the operators onto each basis vector in our space.

If we try the same exercise with the lowering operator, we find that

$$[\hat{S}_z, \hat{S}_-] = -\hbar\hat{S}_-. \tag{3.7}$$

This is similar to the previous one, but has an extra minus sign. Finally, we find that

$$[\hat{S}_+, \hat{S}_-] = 2\hbar\hat{S}_z. \tag{3.8}$$

You are asked to prove these last two commutators in Prob. 3.1.

Note that the commutation relations between these three operators are closed. This means they include only those three operators. This is a property that Lie algebras satisfy, and these three operators from the Lie algebra of rotations, called SU(2) for special unitary group of dimension 2. Lie algebras and Lie groups form an important aspect of much of physics and this is your first example. We will not be going through Lie group properties in detail here, we just wanted to point out the significance of these spin operators.

So far, we have worked with the spin raising, lowering, and z-component operators. Is there another way to organize them, perhaps in a fashion where all the operators act similarly? The answer is yes, when we go to the Cartesian basis, which we do next.

To define Cartesian spin operators we need to find the x- and y-components. These must come from the operators we already have discovered. The relationship is

$$\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-) \text{ and } \hat{S}_y = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-). \tag{3.9}$$

To check that they are somehow "the same", we need to work out their commutation relations. But, now, because we already have the commutation relations in terms of raising, lowering, and z, we can use them to compute the Cartesian commutators. We illustrate with the commutator $[\hat{S}_x, \hat{S}_z]$. We find that

$$[\hat{S}_x, \hat{S}_z] = \frac{1}{2}[\hat{S}_+ + \hat{S}_-, \hat{S}_z] = -\frac{\hbar}{2}(\hat{S}_+ - \hat{S}_-) = -i\hbar\hat{S}_y. \tag{3.10}$$

Repeating for any other pair of Cartesians, we find the result is always $\pm i\hbar$ times the third Cartesian spin operator (this is why the factor was $\hbar$ for the $\hat{S}_\pm$ operator). You should check this with a few other choices to verify that it is true. Summarizing this result gives us

$$[\hat{S}_i, \hat{S}_j] = i\hbar \sum_k \epsilon_{ijk} \hat{S}_k, \tag{3.11}$$

where ϵ_{ijk} is the completely antisymmetric tensor (also called the Levi-Cevita tensor). It is defined by

$$\epsilon_{xyz} = \epsilon_{yzx} = \epsilon_{zxy} = 1, \quad \epsilon_{xzy} = \epsilon_{yxz} = \epsilon_{zyx} = -1, \tag{3.12}$$

and all other possible indices have the tensor vanish: $\epsilon = 0$. You may have seen this tensor either in the definition of the determinant of a 3×3 matrix, or in the cross product between two three-dimensional vectors. There is no physics in this symbol, it just is used to make formulas more compact and easy to write down and read. You get a chance to verify the other five nonzero commutators in Prob. 3.2.

We now need to discuss Hermitian conjugation. Any operator with real eigenvalues that could be measured, such as the Cartesian components of spin, must be Hermitian operators. Hence, $\hat{S}_i^\dagger = \hat{S}_i$. This then tells us that the raising and lowering operators appear in pairs: $\hat{S}_+^\dagger = \hat{S}_-$ and $\hat{S}_-^\dagger = \hat{S}_+$. The Hermitian conjugate of a ket is a bra (when written out in terms of numbers, one takes the transpose to convert a column vector to a row vector and also takes the complex conjugate). To see how a Hermitian conjugate of an operator acts on a bra, we first act on a ket, then take the Hermitian conjugate of both. This procedure is useful in calculations, because we can act operators to the right onto kets or to the left onto bras. One just has to be very careful with complex conjugation when doing this.

You might have been disturbed by the fact that an i appeared in the definition of $\hat{S}_y$. Indeed, it does look odd as to why it would appear. But, it is a natural occurrence coming from the Hermiticity requirement for the Cartesian operators. If we look at Eq. (3.9), we see that because $(\hat{S}_\pm)^\dagger = \hat{S}_\mp$, we can directly verify that both $\hat{S}_x$ and $\hat{S}_y$ are Hermitian. But, the latter is Hermitian only if we have the imaginary factor i! This is why the complex numbers enter into these definitions. Once they are there, the i in the commutator naturally arises as well. So, it should not feel like a mystery to you—instead it is simply required for the theory to be consistent.

Let us recap how measurement worked in the Stern-Gerlach experiment. If we have a state that is up or down in the axis being measured we will always measure it to be up or down. But, if it is a superposition, such as $|\psi\rangle = \cos\alpha\,|\uparrow\rangle - \sin\alpha\,|\downarrow\rangle$ (for an axis in any direction), then the probability to measure up is $\cos^2\alpha$ and the probability to measure down is $\sin^2\alpha$. These were the rules that we introduced, without any clear justification in Chap. 2. Our next goal is to verify the formulas that we previously just told you to use.

This requires us to find the up spin and down spin states for an axis oriented in the θ, ϕ direction in three-dimensions. To find these states, we need to solve an eigenvalue problem, which says

$$\vec{e}_{\theta,\phi} \cdot \hat{\vec{S}}\,|\uparrow\rangle = \frac{\hbar}{2}\,|\uparrow\rangle\,. \tag{3.13}$$

Here, $\vec{e}_{\theta,\phi} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ is the unit vector in the θ, ϕ direction.

Since we have simple formulas for how $\hat{S}_z$, $\hat{S}_+$, and $\hat{S}_-$ act on states, let us try to organize the expression in terms of this. Computing the dot product, we find that

$$\vec{e}_{\theta,\phi} \cdot \hat{\vec{S}} = \sin\theta(\cos\phi\hat{S}_x + \sin\phi\hat{S}_y) + \cos\theta\hat{S}_z = \frac{1}{2}\sin\theta(e^{-i\phi}\hat{S}_+ + e^{i\phi}\hat{S}_-) + \cos\theta\hat{S}_z. \tag{3.14}$$

using this operator, we perform our calculation:

$$\begin{aligned}\vec{e}_{\theta,\phi} \cdot \vec{S}\big(\alpha\,|{\uparrow}\rangle_z + \beta\,|{\downarrow}\rangle_z\big) &= \frac{\hbar}{2}\sin\theta e^{-i\phi}\beta\,|{\uparrow}\rangle_z + \frac{\hbar}{2}\sin\theta e^{i\phi}\alpha\,|{\downarrow}\rangle_z \\ &\quad + \frac{\hbar}{2}\cos\theta\,\alpha\,|{\uparrow}\rangle_z - \frac{\hbar}{2}\cos\theta\,\beta\,|{\downarrow}\rangle_z \\ &= \frac{\hbar}{2}\big(\alpha\,|{\uparrow}\rangle_z + \beta\,|{\downarrow}\rangle_z\big).\end{aligned} \tag{3.15}$$

To solve this, we collect all the coefficients of one of the terms, say $|{\uparrow}\rangle_z$. We find that

$$\frac{\hbar}{2}\sin\theta e^{-i\phi}\beta + \frac{\hbar}{2}\cos\theta\,\alpha = \frac{\hbar}{2}\alpha \tag{3.16}$$

or

$$\frac{\alpha}{\beta} = \frac{\sin\theta}{1-\cos\theta}e^{-i\phi} \tag{3.17}$$

Using the half-angle formulas: $\sin\theta = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}$ and $1-\cos\theta = 2\sin^2\frac{\theta}{2}$, we find that

$$\alpha = \cos\tfrac{\theta}{2} \quad \text{and} \quad \beta = \sin\tfrac{\theta}{2}e^{i\phi}, \tag{3.18}$$

solves the problem for the up-spin projection. A similar analysis for the down spin projection gives

$$\alpha = \sin\tfrac{\theta}{2} \quad \text{and} \quad \beta = -\cos\tfrac{\theta}{2}e^{i\phi}. \tag{3.19}$$

You will verify this second result in Prob. 3.3.

Now, you can see this establishes the rules we used earlier. We had said if the axis is rotated by θ relative to the first, and we take positive projection atoms and place through the second, the probability to obtain a positive projection on the new axis is $\cos^2\frac{\theta}{2}$ and the negative is $\sin^2\frac{\theta}{2}$, which agrees with our calculations. In this way, we have just established the rules we used in Chap. 2.

Our final task is to discuss the total spin operator given by $\hat{\vec{S}}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$. By replacing the x and y-component operators by the raising and lowering ones, we find that

$$\hat{\vec{S}}^2 = \frac{1}{2}\left(\hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+\right) + \hat{S}_z^2. \tag{3.20}$$

Please verify this result before moving on. Using this form, we can operate it on a state. If we operate it on $|{\uparrow}\rangle_z$, we find it is an eigenstate with an eigenvalue of $\frac{3}{4}\hbar^2$. The same holds for the down-spin state.

We have just gone through the operators and states for spin, and we are able to create and work with many different types of operators and states. It is useful to show

how to do some calculations concretely as well, so in Sec. 3.3 we will investigate the representations of these operators as 2×2 matrices named after Wolfgang Pauli. But, before we do that, we need a short tutorial on matrices. Don't worry, everything you need to know follows from dot products of vectors and the quadratic equation.

3.2 Review of matrix properties

A matrix is a group of numbers arranged into an array. We will be working solely with square matrices, where the two dimensions of the array are the same, so $N\times N$ matrices. Matrices satisfy the standard arithmetic operations. I can multiply them by a scalar, add them together if they have the same dimensions (by simply adding each corresponding element "pointwise"). The interesting question comes when we ask how can we multiply them. For this, we will start with matrix-vector multiplication.

When an $N\times N$ matrix multiplies an N-dimensional vector, we do so in a way that involves evaluating N vector-vector dot products. The arrangement is as shown in the equation below for $N=3$:

$$\begin{pmatrix} \boxed{a \quad b \quad c} \\ \boxed{d \quad e \quad f} \\ \boxed{g \quad h \quad i} \end{pmatrix}\begin{pmatrix} \boxed{\begin{matrix} x \\ y \\ z \end{matrix}} \end{pmatrix} = \begin{pmatrix} ax+by+cz \\ dx+ey+fz \\ gx+hy+iz \end{pmatrix} \tag{3.21}$$

The boxes show how we arrange the matrix multiplication, which is, one-by-one, just dot products between a row vector (in the box and coming from the matrix) with the column vector (in the box and on the right).

When we evaluate matrix-matrix multiplication, we simply separate the right matrix into columns and treat each column as a vector, and proceed as before. This leads to the following general formula for the product $C=AB$ of $N\times N$ matrices:

$$C_{mn} = \sum_{l=1}^{N} A_{ml}B_{ln}. \tag{3.22}$$

Be sure this formula makes sense to you, as it is critical for many things we will be doing. If you ever have doubt, recall that the matrix-matrix multiplication is just repeatedly applying matrix vector multiplication to the columns of the second matrix and matrix-vector multiplication is just applying dot products of the rows of the matrix with the vector.

If we can multiply matrices together, the next question arises, is there an identity matrix? The answer is yes, and it is a matrix with 1's along the diagonal and 0s everywhere else. It can be represented with the so-called Kronecker delta function via

$$\mathbb{I}_{mn} = \delta_{mn}, \tag{3.23}$$

where $\delta_{mm} = 1$ and $\delta_{m \neq n} = 0$. Using this result in our general matrix multiplication formula, immediately tells us that $A\mathbb{I} = \mathbb{I}A = A$ indicating it is indeed the identity matrix under multiplication.

Okay, we have an identity matrix, can we find matrix inverses? They would satisfy $AA^{-1} = A^{-1}A = \mathbb{I}$. Here, the answer is that we can find matrix inverses as long as the determinant of the matrix is nonzero. We do not really need to focus on determinants in this book, so we are not going to elaborate for you much about them. For 2×2 and 3×3 matrices, the determinants are

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \quad \text{and} \quad \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh. \tag{3.24}$$

Computing an inverse is not so simple, but we will construct the inverse of a 2×2 matrix. It is

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \tag{3.25}$$

You can see this is well defined only if $\det(A) \neq 0$. Please check and verify that this is the inverse matrix. One can work out the results for larger matrices too, but it gets quite complex quite quickly, so we do not examine anything larger here.

One other curious thing happens when we multiply matrices together. Consider

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}. \tag{3.26}$$

We compute

$$AB = \begin{pmatrix} 1 & 3 \\ -1 & -3 \end{pmatrix} \quad \text{and} \quad BA = \begin{pmatrix} 1 & -1 \\ 3 & -3 \end{pmatrix}. \tag{3.27}$$

You can immediately see that $AB \neq BA$. This often happens with matrices. Multiplication is not commutative. We define the commutator of A and B to be $AB - BA = [A, B]$. When this is nonzero, the two matrices do not commute. The value of the commutator often is important for many quantum-mechanical results, so we will see this concept many times in the remainder of the book. Note further that because we already saw this behavior with the spin operators, that matrices have a chance to represent those operators, because they share this same characteristic.

As you can see from Eq. (3.21), when we multiply a matrix times a vector, we obtain another vector. Since vectors have a magnitude and a direction, the operation corresponding to multiplying a matrix times a vector will generically rotate the vector (to a new direction) and scale it. We call a set of vectors that do not rotate when the matrix multiplies them, they only scale, by the name eigenvector. The scaling factor, which can be negative, is called the eigenvalue. Finding the eigenvectors and eigenvalues of matrices is often an important exercise for many physics problems. There are a number of important facts about this problem. If the matrix we work with is a Hermitian matrix, then we have as many eigenvectors as the dimension of the matrix (so N eigenvectors for an $N \times N$ matrix), the eigenvectors span the

N-dimensional space, are orthogonal to each other, and the eigenvalues are all real. We won't be able to prove any of these results for you. Instead, we will illustrate them explicitly for a 2×2 matrix.

But, before we do this, we need to carefully define what a Hermitian matrix is. It is a matrix that is unchanged when we transpose the matrix elements and take their complex conjugate. In equations, this is

$$(A_{mn})^\dagger = A^*_{nm}. \tag{3.28}$$

Then a Hermitian matrix is a matrix where $A^*_{nm} = A_{mn}$. This means all diagonal elements are real and the off-diagonal elements come in complex-conjugate pairs.

Okay. Let us work out the eigenvalues of a Hermitian 2×2 matrix. We let λ be the eigenvalue, which scales the original vector. So, we have $A\vec{v} = \lambda\vec{v}$, or

$$\sum_l A_{ml} v_n = \lambda v_m. \tag{3.29}$$

We write the right hand side as $\lambda \mathbb{I} v$ and subtract it from both sides of the equation, so that

$$(A - \lambda\mathbb{I})\vec{v} = 0. \tag{3.30}$$

For this equation to have a nonzero solution ($v = 0$ always solves the equation), we require $\det(A - \lambda\mathbf{I}) = 0$. So, let us plug in our Hermitian matrix to calculate this. We have (a and d are real, b can be complex)

$$\det\begin{pmatrix} a-\lambda & b \\ b^* & d-\lambda \end{pmatrix} = (a-\lambda)(d-\lambda) - |b|^2 = 0. \tag{3.31}$$

This can be re-arranged into the quadratic equation

$$\lambda^2 - (a+d)\lambda + ad - |b|^2 = 0, \tag{3.32}$$

with solution

$$\lambda_\pm = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a+d)^2 - 4ad + 4|b|^2} = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a-d)^2 + 4|b|^2}. \tag{3.33}$$

We see that both eigenvalues are real numbers, as we claimed they had to be.

Solving larger eigenvalue problems is a bit more difficult to carry out. But, it follows exactly the same strategy. We form $A - \lambda\mathbb{I}$ and set its determinant to 0. But, solving large problems often requires us to use computers to do so, given the complexity of the mathematics that are involved.

How about the eigenvectors? Well, they are a bit more challenging to determine. One can work out a general formula for the 2×2 problem, but it is a long equation that is not particularly enlightening. So, instead of solving that, we will solve a concrete example.

Find the eigenvalues and eigenvectors of

$$A = \begin{pmatrix} \cos\theta & -i\sin\theta \\ i\sin\theta & -\cos\theta \end{pmatrix}. \tag{3.34}$$

First we obtain the eigenvalues from

$$\lambda^2 - \cos^2\theta - \sin^2\theta = 0 \implies \lambda = \pm 1. \tag{3.35}$$

Then, we pick an eigenvalue, substitute it into our matrix and solve the equation for the eigenvector. Here, we choose $\lambda = 1$:

$$\begin{pmatrix} \cos\theta - 1 & -i\sin\theta \\ i\sin\theta & -\cos\theta - 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0, \quad \text{or} \quad \alpha(\cos\theta - 1) - i\beta\sin\theta = 0, \tag{3.36}$$

where we look only at the top equation. This implies that

$$\frac{\alpha}{\beta} = \frac{i\sin\theta}{1-\cos\theta} = i\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = i\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}. \tag{3.37}$$

It is common to normalize the eigenvectors so that they have length 1 and to have their first nonzero element be real and positive. In this case, we have

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \cos\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} \end{pmatrix}. \tag{3.38}$$

We can repeat this for the $\lambda = -1$ eigenvector, where we would find the eigenvector satisfies

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \sin\frac{\theta}{2} \\ i\cos\frac{\theta}{2} \end{pmatrix}. \tag{3.39}$$

You can verify that these two eigenvectors are indeed orthogonal (do not forget the complex conjugation for the row vector). It takes some practice, but you should get the hang of it after trying some examples.

Matrices, like vectors, can be expressed in many different bases. If the matrix is Hermitian, one useful basis is the eigenvector basis. In this basis, the matrix is diagonal with the corresponding eigenvalues occupying each diagonal element.

One can also examine functions of matrices. The simplest way to think of this is to expand the function in a power series and thereby determine the function of the matrix as a power series of the matrix. There are questions of convergence of the series if we do that, though. So, a second way is to first diagonalize the matrix, then evaluate the function in the diagonal basis. Since the product of a diagonal matrix with itself remains diagonal, one immediately finds that

$$f(A) = \mathrm{Diag}(f(\lambda_1), f(\lambda_2), \cdots, f(\lambda_n)), \tag{3.40}$$

where Diag is an operation that places the N elements, in order, onto the diagonal elements of a matrix and the eigenvalues are $\lambda_1, \lambda_2, \cdots, \lambda_N$. When we use this approach, we do not need the function to be expandable in a power series. So, for example, we can compute the square root of a matrix this way. Of course, one of the most interesting examples is the exponential function. We will explore calculating the exponentials of certain 2×2 matrices later.

We end by defining one more type of matrix, the unitary matrix. It has its Hermitian conjugate equal to its inverse. So $UU^\dagger = U^\dagger U = \mathbb{I}$. Interestingly, unitary matrices can always be written in the form $U = \exp(iH)$, where H is a Hermitian matrix. One then sees that $U^\dagger = \exp(-iH^\dagger) = \exp(-iH)$, which obviously is the inverse of $\exp(iH)$.

3.3 Pauli spin matrices

Okay, we had a long mathematical discussion about matrices and we summarized a number of their properties. Now, we wish to apply these ideas to the representation of the spin operators as 2×2 matrices, which involves using the so-called Pauli spin matrices, named after Wolfgang Pauli, who invented them [1]. Pauli is depicted in Fig. 3.1 from about the time he discovered these matrices. Pauli was a German physicist and he has a lot of stories about him. He was a child prodigy, who learned general relativity when he was in his teens. He wrote a review article on it that remained the gold standard for the field for many years. He tended to eschew more mathematical treatments and always insisted on focusing on the physical side of descriptions. He is also known to be very unlucky to have in the lab. It seemed like whenever he visited one, equipment would just stop working.

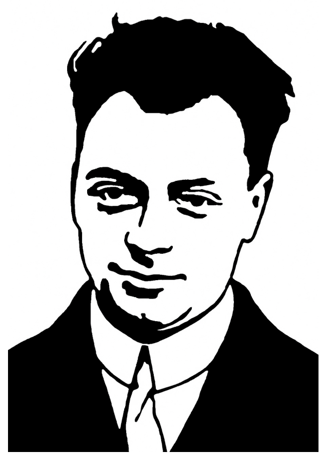

Fig. 3.1: Wolfgang Pauli in 1926. Pauli was an outstanding theoretical physicist who determined numerous important results including the Pauli exclusion principle and the CPT theorem. He also discovered the Pauli spin matrices. [Credit. Public domain. Image converted to look like a stencil by the authors.]

To discover the Pauli spin matrices, we first have to carefully go through the procedure we must follow to create the matrix representation of an operator. Then, we can apply the procedure to the spin operators.

The procedure is a cookbook. Just follow the directions and all will be well. We start with an orthogonal basis for our vector space $\{|\psi_n\rangle \ : \ \forall\ n \text{ with } 1 \le n \le N\}$ which is also normalized (called an orthonormal basis). Then we act the operator $\hat{O}$ onto the first vector: $\hat{O}\,|\psi_1\rangle$. This is a vector in our space, so it can be expanded in terms of our basis vectors. Hence, $\hat{O}\,|\psi_1\rangle = \sum_{m=1}^{N} O_{m1}\,|\psi_m\rangle$, where $O_{m1} = \langle\psi_m|\hat{O}|\psi_1\rangle$. This produces the first column of the matrix representation of $\hat{O}$. We then proceed through the remaining $N-1$ basis vectors to compute the full matrix.

Let us do this for the spin operators. We have $\hat{S}_z\,|\uparrow\rangle_z = \frac{\hbar}{2}|\,|\uparrow\rangle_z$ and $\hat{S}_z\,|\downarrow\rangle_z = -\frac{\hbar}{2}\,|\downarrow\rangle_z$, so that

$$\hat{S}_z \leftrightarrow \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = \frac{\hbar}{2}\sigma_z, \tag{3.41}$$

which defines the σ_z Pauli matrix.

We now obtain the other two. We have $\hat{S}_x\,|\uparrow\rangle_z = \frac{\hbar}{2}\,|\downarrow\rangle$ and $\hat{S}_x\,|\downarrow\rangle_z = \frac{\hbar}{2}\,|\uparrow\rangle_z$, so that

$$\hat{S}_y \leftrightarrow \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \frac{\hbar}{2}\sigma_x. \tag{3.42}$$

And finally, we also have $\hat{S}_y\,|\uparrow\rangle_z = \frac{i\hbar}{2}\,|\downarrow\rangle_z$ and $\hat{S}_y\,|\downarrow\rangle_z = -\frac{i\hbar}{2}\,|\uparrow\rangle_z$. Hence,

$$\hat{S}_y \leftrightarrow \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} = \frac{\hbar}{2}\sigma_y. \tag{3.43}$$

The spin operators satisfy a commutation relation $[\hat{S}_i, \hat{S}_j] = i\hbar\sum_k \epsilon_{ijk}\hat{S}_k$, which implies the matrices should satisfy a similar relationship, given by

$$[\sigma_i, \sigma_j] = 2i\sum_k \epsilon_{ijk}\sigma_k. \tag{3.44}$$

Let us verify this for $i = x$ and $j = y$. We have that

$$\sigma_x\sigma_y = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} = \begin{pmatrix}i & 0\\ 0 & -i\end{pmatrix} = i\sigma_z \ \text{ and } \ \sigma_y\sigma_x = \begin{pmatrix}-i & 0\\ 0 & i\end{pmatrix} = -i\sigma_z. \tag{3.45}$$

Hence, we do see that $[\sigma_x, \sigma_y] = 2i\sigma_z$, as we claimed it should. In Prob. 3.7, you will verify the other two commutation relations.

We also have an interesting surprise. If we look at the anticommutator, defined by $\{A, B\}_+ = AB + BA$, we find that

$$\{\sigma_x, \sigma_y\}_+ = i\sigma_z - i\sigma_z = 0. \tag{3.46}$$

This means the Pauli spin matrices anticommute. In Prob. 3.8, you will verify that $\{\sigma_y, \sigma_z\}_+ = 0$ and $\{\sigma_z, \sigma_x\}_+ = 0$. This property is not expected from the commutation relations of spin. Finally, we also have that each Pauli spin matrix squares to one: $\sigma_i^2 = \mathbb{I}$. Note that these matrices are both Hermitian and unitary at the same time, which is a property that does not often come together.

3.4 Pauli spin matrix identities

We have already shown that the Pauli spin matrices satisfy a commutation relation given by $[\sigma_i, \sigma_j] = 2i \sum_k \epsilon_{ijk}\sigma_k$ and an anticommutation relation $\{\sigma_i, \sigma_j\}_+ = 2\mathbb{I}\delta_{ij}$. We can use these two together to derive the product formula for two Pauli matrices, which is derived as follows:

$$\sigma_i \sigma_j = \tfrac{1}{2}\{\sigma_i, \sigma_j\}_+ + \tfrac{1}{2}[\sigma_i, \sigma_j] = \mathbb{I}\delta_{ij} + i \sum_k \epsilon_{ijk}\sigma_k. \tag{3.47}$$

This product formula can be very helpful in simplifying complicated formulas involving Pauli spin matrices. Related to this formula, we have the triple product formula, which says

$$\sigma_x \sigma_y \sigma_z = i\mathbb{I}. \tag{3.48}$$

While this result is often not so useful, it is cute to see.

Another important notation is a shorthand identity for a 2×2 matrix constructed out of Pauli spin matrices. We use a dot product to do this, so

$$\vec{v} \cdot \vec{\sigma} = v_x \sigma_x + v_y \sigma_y + v_z \sigma_z = \begin{pmatrix} v_z & v_x - iv_y \\ v_x + iv_y & -v_z \end{pmatrix}. \tag{3.49}$$

Students often can get confused by this notation, so be sure you can follow it. The dot product is the sum of three matrices weighted by real coefficients.

We can study the eigenvalues of this matrix. Recall our procedure: we compute $\det(\vec{v} \cdot \vec{\sigma} - \lambda\mathbb{I})$, which becomes $\lambda^2 - \|\vec{v}\|^2 = 0$. Hence, the eigenvalues of $\vec{v} \cdot \vec{\sigma}$ are given by $\pm\|\vec{v}\|$. This can allow you to read off the eigenvalues if you can identify v_x, v_y, and v_z from a matrix. If the matrix has any component of the identity in it, you would just add that component to the eigenvalues we already found. This is a great shortcut to finding eigenvalues of 2×2 matrices.

Previously, in Sec. 3.1, we showed how to find the states with positive and negative projections of spin onto an axis in the θ, ϕ direction. Here, we will redo this calculation using the matrix representations. In the end, you can decide which approach you prefer for yourself.

Recall, we worked with the operator $\vec{e}_{\theta,\phi} \cdot \hat{\vec{S}}$, which is represented by

$$\vec{e}_{\theta,\phi} \cdot \hat{\vec{S}} \leftrightarrow \frac{\hbar}{2}\vec{e}_{\theta,\phi} \cdot \vec{\sigma} = \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}. \tag{3.50}$$

Our goal is to find the eigenvalues and eigenvectors. I hope you can immediately read off that the eigenvalues are $\pm\frac{\hbar}{2}$ by using our trick we discussed above (if not, you can calculate the appropriate determinant, and solve for the two roots).

Now to get the positive eigenvector, we need to solve

$$\begin{pmatrix} \cos\theta - 1 & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta - 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \tag{3.51}$$

and make sure that $|\alpha|^2 + |\beta|^2 = 1$. Looking at the top equation, we find that $(\cos\theta - 1)\alpha + \sin\theta e^{-i\phi}\beta = 0$, which is solved by

$$\frac{\alpha}{\beta} = \frac{\sin\theta e^{-i\phi}}{1 - \cos\theta} = \frac{2\cos\frac{\theta}{2}\sin\frac{\theta}{2}e^{-i\phi}}{2\sin^2\frac{\theta}{2}} = \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}e^{i\phi}}, \tag{3.52}$$

which suggests the choice $\alpha = \cos\frac{\theta}{2}$ and $\beta = \sin\frac{\theta}{2}e^{i\phi}$. Note that this is the same answer as we obtained before in Eq. (3.18). Now, there is a trick to get the down-spin result. Because the eigenvectors are orthogonal to each other, we simply need to find the orthogonal vector to (α, β). Recalling the need for the complex conjugate, we find this is $(\beta^*, -\alpha^*)$, which gives $\alpha = \sin\frac{\theta}{2}e^{-i\phi}$ and $\beta = -\cos\frac{\theta}{2}$ for the down projection eigenvector. But, we usually want the first entry to be real, so we multiply the eigenvector by $e^{i\phi}$, a complex phase, and get the final result of $\alpha = \sin\frac{\theta}{2}$ and $\beta = -\cos\frac{\theta}{2}e^{i\phi}$.

Now, many students, especially those who have seen some linear algebra, prefer doing calculations concretely this way with matrices. But, we want to encourage you to always remember the abstract way of doing things, with operators and states, because we sometimes cannot do these problems using matrices.

We have one more identity to work out and it comes from our product formula in Eq. (3.47). Because the Levi-Cevita tensor is antisymmetric under the interchange of any two of its indices, we have that

$$(\vec{v}\cdot\vec{\sigma})^2 = \sum_{ij} v_i v_j \sigma_i \sigma_j = \sum_{ij} v_i v_j \left(\mathbb{I}\delta_{ij} + i\sum_k \epsilon_{ijk}\sigma_k\right) = \|\vec{v}\|^2\mathbb{I}, \tag{3.53}$$

because $\sum_{ij} v_i v_j \epsilon_{ijk} = 0$, which you can verify by interchanging i and j and seeing it is the negative of this. If you are having trouble understanding this, just write out the result completely, and you will see the cancellation.

We can use this to work out a generalized Euler identity, which computes $e^{i\vec{v}\cdot\vec{\sigma}}$, generalizing the conventional Euler identity to these 2×2 matrices. We do this by writing the exponential as a power series, and using the identity we just worked out. We have

$$e^{i\vec{v}\cdot\vec{\sigma}} = \sum_{n=0}^{\infty}\frac{i^n}{n!}(\vec{v}\cdot\vec{\sigma})^n = \sum_{n=0}^{\infty}\frac{(-1)^n(\|\vec{v}\|^2)^n}{(2n)!}\mathbb{I} + \sum_{n=0}^{\infty}\frac{(-1)^n(\|\vec{v}\|^2)^n}{(2n+1)!}i\vec{v}\cdot\vec{\sigma}, \tag{3.54}$$

which comes from separating out the odd and even power terms, and using the fact that $(\vec{v}\cdot\vec{\sigma})^{2n} = (\|\vec{v}\|^2)^n\mathbb{I}$. Now, recognizing the power series for cosine and sine, we find that

$$e^{i\vec{v}\cdot\vec{\sigma}} = \cos(\|\vec{v}\|)\mathbb{I} + i\sin(\|\vec{v}\|)\frac{\vec{v}\cdot\vec{\sigma}}{\|\vec{v}\|}, \tag{3.55}$$

which is our generalized Euler identity for these 2×2 matrices.

We can use our generalized Euler identity to show that, in general $e^{i\vec{v}\cdot\vec{\sigma}}$ and $e^{i\vec{v}\,'\cdot\vec{\sigma}}$ do not commute. For, we have that

$$\begin{aligned}
e^{i\vec{v}\cdot\vec{\sigma}}e^{i\vec{v}\,'\cdot\vec{\sigma}} &= \left(\cos(\|\vec{v}\|)\mathbb{I} + i\sin(\|\vec{v}\|)\frac{\vec{v}\cdot\vec{\sigma}}{\|\vec{v}\|}\right)\left(\cos(\|\vec{v}\,'\|)\mathbb{I} + i\sin(\|\vec{v}\,'\|)\frac{\vec{v}\,'\cdot\vec{\sigma}}{\|\vec{v}\,'\|}\right)\\
&= \cos(\|\vec{v}\|)\cos(\|\vec{v}\,'\|)\mathbb{I} + i\cos(\|\vec{v}\|)\sin(\|\vec{v}\,'\|)\frac{\vec{v}\,'\cdot\vec{\sigma}}{\|\vec{v}\,'\|} + i\cos(\|\vec{v}\,'\|)\sin(\|\vec{v}\|)\frac{\vec{v}\cdot\vec{\sigma}}{\|\vec{v}\|}\\
&\quad - \sin(\|\vec{v}\|)\sin\|\vec{v}\,'\|)\frac{\vec{v}\cdot\vec{\sigma}\,\vec{v}\,'\cdot\vec{\sigma}}{\|\vec{v}\|\,\|\vec{v}\,'\|}\\
&= \cos(\|\vec{v}\|)\cos(\|\vec{v}\,'\|)\mathbb{I} + i\cos(\|\vec{v}\|)\sin(\|\vec{v}\,'\|)\frac{\vec{v}\,'\cdot\vec{\sigma}}{\|\vec{v}\,'\|} + i\cos(\|\vec{v}\,'\|)\sin(\|\vec{v}\|)\frac{\vec{v}\cdot\vec{\sigma}}{\|\vec{v}\|}\\
&\quad - \sin(\|\vec{v}\|)\sin(\|\vec{v}\,'\|)\frac{\vec{v}\cdot\vec{v}\,'}{\|\vec{v}\|\,\|\vec{v}\,'\|}\mathbb{I} - i\sin(\|\vec{v}\|)\sin\|\vec{v}\,'\|)\frac{\vec{\sigma}\cdot(\vec{v}\times\vec{v}\,')}{\|\vec{v}\|\,\|\vec{v}\,'\|},
\end{aligned} \tag{3.56}$$

where we used the Pauli spin matrix product formula in Eq. (3.47) to simplify the last part.

It is the cross-product term that changes sign if we interchange the orders of the two exponentials. So we learn that these two exponentials do not commute if the two vectors in each exponent are not parallel. While one might have been able to guess such a rule, proving it is a bit of a challenge.

3.5 Application of the formalism to the Stern-Gerlach experiment

We have developed a lot of formalism about spin and the Pauli spin matrices. We need to do some physics. This is our next task. A good rule of thumb, which we will see in more detail when we discuss quantum computing, is that one can always think of a quantum experiment as a state preparation followed by a measurement. Using our Dirac notation and applying operators to states prior to the measurement is how we perform the state preparation. We can, if we want to, convert the abstract operators and states into matrices and vectors and use matrix manipulations to carry out the state preparation. In this section, we will describe a few ways that one can approach such problems.

Let us discuss how we can simulate the boiling off of silver atoms from an oven, their entrance into a Stern-Gerlach analyzer, and their exit from the analyzer. Since the oven creates random quantum states of spin, we would randomly select

ϕ, randomly select $\cos\theta$, and randomly choose a positive or negative projection. Then, we know exactly what state is input into the analyzer. We then let it spatially separate and exit each exit. We finally have to measure it. How would this work in the Dirac notation? Well, suppose we work with the positive projection, then we start in the state $\cos\frac{\theta}{2}\,|\uparrow\rangle_z + \sin\frac{\theta}{2}e^{i\phi}\,|\downarrow\rangle_z$. If we choose to describe the system with just the spin degree of freedom, then the state remains the same after traveling through the device. It is only at the end, where we measure those that are in the $|\uparrow\rangle_z$ or $|\downarrow\rangle_z$ states that we determine the probabilities to emerge, which are $\cos^2\frac{\theta}{2}$ for the positive projection and $\sin^2\frac{\theta}{2}$ for the negative projection. Note that this way of thinking about the problem does not really have much of a state preparation step to it, as the initial state remains the same state throughout until the measurement. Of course this works, but it somehow does not seem to be right.

So, we can do better. We can say that the analyzer correlates the spin degree of freedom with the spatial degree of freedom (and it really does do this), which would create an entangled state. Then, we would actually be measuring the position at the end and not the spin, because we just collect the atoms at either of the two spatial positions corresponding to the locations of the exits. Indeed, this is beginning to sound more like the real experiment. How does this analysis go?

We start with the same state, except now we have an additional spatial degree of freedom, which we call beam, because we start inside the beam. Hence, our initial state is

$$\cos\frac{\theta}{2}\,|\uparrow\rangle_z \otimes |\text{beam}\rangle + \sin\frac{\theta}{2}e^{i\theta}\,|\downarrow\rangle_z \otimes |\text{beam}\rangle\,. \tag{3.57}$$

We have introduced a funny notation called the tensor product sign $\otimes$. We use this to separate the states of different character, or of different degrees of freedom (here, we have spin and space). The tensor product allows us to keep track of these different degrees of freedom. We can have operators act on either space separately, or on both together. But a spin operator can only act on a spin state and a spatial operator can only act on a spatial state.

Okay. Now, we go through the analyzer which separates the atoms according to the spin. Our state now becomes

$$\cos\frac{\theta}{2}\,|\uparrow\rangle_z \otimes |\text{up exit}\rangle + \sin\frac{\theta}{2}e^{i\theta}\,|\downarrow\rangle_z \otimes |\text{down exit}\rangle\,. \tag{3.58}$$

What operator can make such a change? It is the following:

$$\hat{O} = |\uparrow\rangle_z\,{}_z\langle\uparrow| \otimes |\text{up exit}\rangle\,\langle\text{beam}| + |\downarrow\rangle_z\,{}_z\langle\downarrow| \otimes |\text{down exit}\rangle\,\langle\text{beam}|\,. \tag{3.59}$$

Now, when this acts on our tensor-product state, the operators act on the states they are allowed to, as indicated by the tensor-product sign. Then, if we note that the spin states are orthogonal and normalized, and that the beam state is normalized, we obtain the result we showed above. Please go through this exercise yourself to be sure you understand how this works. It takes some practice to work with these tensor product states and operators, but you can see it is a pretty straightforward process. Now, after preparing the state, we measure them. But, what we really measure is just the spatial state, and we infer the spin because we constructed this particular state that correlates the position with the spin. This state is called an entangled state, because we cannot factorize it across the tensor-product symbol. Whereas, the initial state can have $|\text{beam}\rangle$ factored out of both components—we call such a state a product state.

What this means is that we can view the Stern-Gerlach analyzer as creating an entangled state that allows us to infer the spin of the atom, which we cannot see, or easily measure, with the spatial location of the atom, which we can measure, albeit with difficultly. It is all the same experiment, but you see, by performing a more careful analysis, we have a deeper understanding about how it works and about what states are actually being prepared.

Now, let us move onto the analyzer-loop experiments. Let us start with the simplest case—a positive-z source and a positive-x-oriented analyzer loop, as in Fig. 2.18. As we go through the experiment we have the following evolution. We start in the state $|\uparrow\rangle_z \otimes |\text{beam}\rangle$. Then, as we separate in the x-oriented loop, we create the state $\frac{1}{\sqrt{2}}\left(|\uparrow\rangle_x \otimes |\text{front}\rangle + |\downarrow\rangle_x \otimes |\text{back}\rangle\right)$. Note that this is an entangled state. If we do nothing, we will go back to the original state $|\uparrow\rangle_z \otimes |\text{beam}\rangle$. But, if we do something, like measure on the front arm with a pass-through detector, then we create the state $\frac{1}{\sqrt{2}}|\uparrow\rangle_z \otimes |\text{beam}\rangle$ upon the exit, if the detector fired, and the state $\frac{1}{\sqrt{2}}|\downarrow\rangle_x \otimes |\text{beam}\rangle$, if it did not fire. Note that we obtain these states by applying the projection operator corresponding to the measurement on the different arms (corresponding to $|\uparrow\rangle_x\ {}_x\langle\uparrow|$ or $|\downarrow\rangle_x\ {}_x\langle\downarrow|$, respectively), and then evolving the position state in the arms into the beam state upon exit. We can construct the operators that do this, but we hold off on that at the moment.

This approach, of working with the state preparation and Dirac notation takes nothing away from the previous way that we analyzed the problems using our rules. Both are perfectly valid and contain essentially the same information. In Chap. 4, we will explore similar ways of describing experiments with light. Our rules-based method will use a model developed by Richard Feynman that is a simplified version of his famous path-integral approach to quantum mechanics. The Dirac-notation-based approach will be similar to what we did here, but of course will have a number of differences as well.

3.6 Summary

We did a lot in this chapter! We started from the experimental observations of the Stern-Gerlach analyzer and showed how one can use those results to develop an abstract state and operator-based approach. We learned how to determine the commutation relations of these operators from their definitions and we learned how to create states with definite projections of spin in any direction. We also showed a more concrete approach that worked with matrices and vectors that represent the operators and states using a specific set of basis vectors. The matrices that represent spin, called the Pauli spin matrices, are particularly important and we investigated a number of their properties including two identities—the Pauli spin matrix product identity and the generalized Euler identity. We saw that we could analyze a few interesting questions about the operators by working with these identities.

We now move on to discuss light.

Chapter 3 problems

Problem 3.1 Spin commutation relations. Prove the two commutators in Eqs. (3.7) and (3.8).

Problem 3.2 Cartesian spin commutation relations. Compute $[\hat{S}_i, \hat{S}_j]$ for $ij = xy$, yx, yz, zy, xz and zx. Verify that the Cartesian commutator obeys the relation in Eq. (3.11).

Problem 3.3 Negative-projection spin state in an arbitrary direction. Finish the calculation for the state of definite spin that has a down-spin projection in the θ, ϕ direction.

Problem 3.4 Total spin is a good quantum number. Show that $\hat{\vec{S}}^2$ commutes with all three Cartesian component spin operators, or $[\hat{\vec{S}}^2, \hat{S}_i] = 0$.

Problem 3.5 Practice with eigenvalue-eigenvector problems. Find the eigenvalues and eigenvectors of

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} \tag{3.60}$$

and verify that the eigenvectors you find are orthogonal. Remember to take the complex conjugate of the row vector in the inner product!

Problem 3.6 Nonuniqueness of the eigenvector. We said that an eigenvector satisfies $A\vec{v} = \lambda\vec{v}$. If we consider a new vector $\vec{v}' = \alpha\vec{v}$ with $\alpha \neq 0$. Is $\vec{v}'$ also an eigenvector? What eigenvalue does it have if it is an eigenvector? If this works and we insist on the eigenvectors being normalized, then $\alpha = e^{i\theta}$ must be a complex phase that lies on the unit circle in the complex plane. Explain how this can be used to guarantee the first nonzero element of an eigenvector can always be made to be real and positive.

Problem 3.7 Commutation relation for Pauli spin matrices. Verify the two commutation relations $[\sigma_y, \sigma_z] = 2i\sigma_x$ and $[\sigma_z, \sigma_x] = 2i\sigma_y$ by explicitly multiplying out the matrices.

Problem 3.8 Anticommutation of Pauli spin matrices. Verify that $\{\sigma_y, \sigma_z\}_+ = 0$ and $\{\sigma_z, \sigma_x\}_+ = 0$.

Problem 3.9 Properties of Pauli spin matrices. Verify that the Pauli spin matrices satisfy all of the following properties: (i) they are Hermitian; (ii) they are unitary;

(iii) they have a determinant equal to one; and (iv) they have a trace equal to zero. Note that the trace satisfies $\mathrm{Tr}A = \sum_{n=1}^{N} A_{nn}$, that is, it sums all of the diagonal elements of the matrix.

Suppose we diagonalized a Hermitian 2×2 matrix so it takes the form with λ_1 on the top diagonal and λ_2 on the bottom diagonal. Calculate the determinant and trace in terms of λ_i.

Problem 3.10 Raising and lowering Pauli spin matrices. Determine what the corresponding Pauli raising matrix σ_+ and lowering matrix σ_- are.

Problem 3.11 Practice with state preparation and tensor products. In the text, we discussed the state preparation in an analyzer loop where a pass through detector was placed on the front arm only. By adding a new tensor product degree of freedom, corresponding to $|\text{ready}\rangle$ to indicate the pass through detector is ready to detect and $|\text{fired}\rangle$ to indicate that it fired to detect a particle, re-analyze the experiment we worked with before, but now with the three-fold tensor product basis of the form $|\text{spin}\rangle \otimes |\text{position}\rangle \otimes |\text{detector}\rangle$. Is the final state that you create upon exiting the analyzer loop entangled, partially entangled, or a product state? Note that a partially entangled state will factorize with respect to one, but not two degrees of freedom for this state.

References

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